

When and How to Pilot: Design Rules for Two-Wave Experiments

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Platform experiments often begin with a ramp-up, a first wave run on a small fraction of traffic before the full experiment [Kohavi et al., 2020]. The data from that first wave can inform a consequential design choice, the traffic split between treatment and control in the main wave. This choice determines how many observations each arm gets and hence how precisely the average treatment effect is estimated. In particular, assigning more observations to the noisier arm improves precision without increasing the total sample size. If the first wave is small, however, its data say little about which arm is noisier, and reallocating traffic on that basis can leave the main wave worse off than a simple 50/50 split.

This paper asks how much the main-wave allocation should respond to the evidence in a small pilot. The two common answers are to ignore the pilot and keep a 50/50 split, or to take its variance estimates as exact and allocate accordingly. I propose a middle ground. I build a range of plausible values for each arm’s variance from the pilot and pick the split that performs best in the worst case over those values. When the pilot says little, the ranges are wide and the design stays close to 50/50. As it says more, they narrow and the design moves toward the optimal split. The chosen split also comes with a guarantee, computed before the main wave runs, on how much precision it can lose. Simulations calibrated to two platform experiments show that this middle ground avoids the severe losses of trusting the estimates when the pilot is small, while matching their gains once it is large.

1 The two-wave allocation problem

To formalize the problem, I assume that the pilot and the main wave are independent samples from the same population, and that potential outcomes lie in $[0, 1]$, which bounds each arm’s variance by $1/4$. The experimenter runs a pilot of total size M , split equally between treatment and control, and then fixes the treatment assignment probability for the main wave. The goal is to estimate the average treatment effect (ATE) in the main wave as precisely as possible.

Assigning a share π of the main wave to treatment gives the difference-in-means estimator of the ATE a variance proportional to

$$V(\pi, \sigma_1^2, \sigma_0^2) = \frac{\sigma_1^2}{\pi} + \frac{\sigma_0^2}{1 - \pi},$$

where σ_1^2 and σ_0^2 are the variances of the potential outcomes under treatment and control. If these were known, the experimenter could use the Neyman allocation [Neyman, 1934], $\pi^*(\sigma_1^2, \sigma_0^2) = \sigma_1/(\sigma_1 + \sigma_0)$, which minimizes V at $V^*(\sigma_1^2, \sigma_0^2) = (\sigma_1 + \sigma_0)^2$. I measure the cost of any split π by its regret, the excess variance relative to the Neyman allocation,

$$r(\pi, \sigma_1^2, \sigma_0^2) = V(\pi, \sigma_1^2, \sigma_0^2) - V^*(\sigma_1^2, \sigma_0^2) = (\sigma_1 + \sigma_0)^2 \frac{(\pi - \pi^*)^2}{\pi(1 - \pi)},$$

which is zero at the Neyman allocation and grows with the squared distance from it.

In practice, of course, σ_1^2 and σ_0^2 are unknown when the main-wave design is chosen. One common approach is to keep the main wave balanced, sending half the sample to each arm regardless of the pilot. Although it does not adapt to the data, balanced assignment has a decision-theoretic justification. It is the minimax-risk rule, the assignment with the smallest worst-case variance of the ATE estimator. Without a pilot, balance is also the unique minimax-regret rule, with a worst-case regret of $1/4$. Balance can nevertheless be costly when the variances differ. Specifically, to match

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the precision of the Neyman allocation, it needs $(\sigma_1 - \sigma_0)^2 / (\sigma_1 + \sigma_0)^2$ more observations, that is, 11 percent more when one standard deviation is twice the other and 25 percent more when it is three times as large.

The other common approach is the feasible Neyman allocation (FNA), which estimates σ_1^2 and σ_0^2 from the pilot and plugs them into the Neyman formula. With a large pilot, FNA is asymptotically efficient [Hahn et al., 2011, Armstrong, 2026]. With a small pilot, however, the estimates are noisy, and nothing limits how far they can tilt the assignment. An arm whose pilot variance comes out small receives almost no main-wave traffic, and if that arm is in fact noisy, the few observations it gets leave the ATE estimate far less precise than under balance [Cai and Rafi, 2024]. Formally, the worst-case regret of FNA is unbounded at every pilot size. A minimum share per arm, a common fix, keeps this loss finite but does not make it small. Balance, in short, never moves, and FNA can move too far.

How far should the assignment move between these two approaches? The answer from statistical decision theory is the minimax regret rule, the rule with the smallest worst-case expected regret across all populations the experimenter might face. This rule does use the pilot, since with as few as two observations per arm its worst-case expected regret falls below the no-pilot level of $1/4$. Moreover, as the pilot grows, its worst-case expected regret falls at the $M^{-1/2}$ rate. Computing it, however, requires a complete plan of what to do for every pilot the experimenter might see, and no tractable way to find that plan is known. The rule I propose keeps the minimax-regret logic but uses it only after the pilot has been seen.

2 Conditional minimax regret

I choose the main-wave split to minimize worst-case regret over the variance pairs that the pilot has not ruled out. To define those pairs, I construct a confidence interval for each arm’s variance. This gives the Conditional Minimax Regret (CMR) rule, in the spirit of the decision making with set estimates discussed by Manski [2021]. With a small pilot the confidence intervals are wide, the worst case is nearly as bad as with no pilot, and the chosen split stays close to balance. With a large pilot the confidence intervals collapse around the estimates and the split approaches the Neyman allocation.

The two intervals form a rectangle $\hat{\Theta}_\alpha \subseteq [0, 1/4]^2$ that contains (σ_1^2, σ_0^2) with probability at least $1 - \alpha$. To construct the intervals, I use the bounds of Maurer and Pontil [2009], which provide finite-sample coverage for every outcome distribution on $[0, 1]$. For binary outcomes, I instead use exact finite-sample inversion, which gives tighter intervals. CMR minimizes worst-case regret over this rectangle,

$$p_{\text{CMR}} \in \arg \min_{\pi} \sup_{(\sigma_1^2, \sigma_0^2) \in \hat{\Theta}_\alpha} r(\pi, \sigma_1^2, \sigma_0^2).$$

With $\underline{\sigma}_d$ and $\bar{\sigma}_d$ the lower and upper bounds of the confidence interval for the standard deviation of arm d , the minimizer has a closed form,

$$p_{\text{CMR}} = \frac{(\underline{\sigma}_1 + \bar{\sigma}_1)/2}{(\underline{\sigma}_1 + \bar{\sigma}_1)/2 + (\underline{\sigma}_0 + \bar{\sigma}_0)/2},$$

i.e., the Neyman formula evaluated at the midpoint of each arm’s interval. CMR also reports a certificate, U_{CMR} , the largest regret the chosen split can incur anywhere in the rectangle [Andrews and Chen, 2025]. Figure 1 illustrates the allocation and the certificate for three stylized rectangles. With no pilot the split is 50/50 and the certificate is $1/4$, a small pilot moves them to 64/36 and 0.069, and a larger pilot to 73/27 and 0.013, close to the Neyman split of 75/25.

The rule has two attractive properties, one that holds for every pilot size and one that holds as the pilot grows. First, for every pilot size, the certificate bounds the regret of the chosen split with probability at least $1 - \alpha$ and never exceeds the certificate that balance would report.

Theorem 1 (The protection of balance when the pilot is small). *For every outcome distribution with bounded support and every pilot size,*

- (i) $\Pr\{r(p_{\text{CMR}}, \sigma_1^2, \sigma_0^2) \leq U_{\text{CMR}}\} \geq 1 - \alpha$.
- (ii) $U_{\text{CMR}} \leq 1/4$, the worst-case regret of balance, and $U_{\text{CMR}} \rightarrow 0$ as the pilot grows.

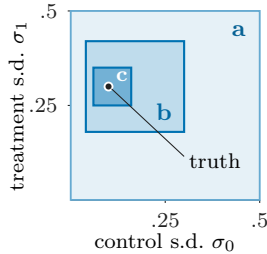
The rectangle contains the true variances with probability at least $1 - \alpha$, and whenever it does, the certificate bounds the regret of the chosen split, which gives part (i). For part (ii), CMR picks the split with the smallest worst-case regret over the rectangle, and balance is one of the candidates. The certificate therefore cannot exceed balance’s worst-case regret, which is at most $1/4$ for any rectangle. As the pilot grows, the rectangle contracts around the true variances and the certificate converges to zero.

Second, CMR learns the Neyman allocation as fast as FNA once the pilot is large.

Theorem 2 (The efficiency of feasible Neyman when the pilot is large). *As the pilot grows,*

- (i) *If both variances are positive, p_{CMR} converges to the Neyman allocation at rate $M^{-1/2}$ and its regret vanishes at rate M^{-1} , the same rates as FNA.*
- (ii) *Over all populations with bounded outcomes, the worst-case expected regret of p_{CMR} is of order $M^{-1/2}$, the best rate any pilot-based rule can achieve, up to a constant that depends only on α .*

Part (i) holds because the confidence intervals shrink at the $M^{-1/2}$ rate as the pilot grows, and the CMR split, which sits at their midpoints, approaches the Neyman allocation at the same rate. Since regret is proportional to the squared distance $(\pi - \pi^*)^2$, it vanishes at the M^{-1} rate. Part (ii) says that CMR performs as well as the minimax regret rule in the worst case, up to a constant. Taken together, the two theorems deliver the protection of balance when the pilot is small and the efficiency of FNA when it is large.



	σ_1 interval	σ_0 interval	p_{CMR}	U_{CMR}
(a) no information	[0, .50]	[0, .50]	0.50	0.25
(b) small pilot	[.18, .42]	[.04, .30]	0.64	0.069
(c) larger pilot	[.25, .35]	[.06, .16]	0.73	0.013
truth	.30	.10	$\pi^* = 0.75$	0

Figure 1: Three stylized confidence rectangles and the CMR assignment implied by each. Rectangle (a) is the full range $[0, 1/2]$ of each arm’s standard deviation, i.e., the case of no pilot. Rectangles (b) and (c) are not realized confidence sets but illustrations of what a small and a larger pilot might produce when the truth is $\sigma_1 = 0.30$ and $\sigma_0 = 0.10$, for which the Neyman allocation is 0.75.

3 Shared-control and stratified designs

CMR’s logic extends to other common experimental designs. Table 1 sets two of them beside the two-arm case. In an A/B/n test, K treatments share one control group, which enters every treatment-effect estimate. The loss is then the sum of the variances of the K treatment-control contrasts, and because the control arm enters every contrast, its variance counts K times. The Neyman allocation gives it $\sqrt{K}$ times the share of a treatment with an equal standard deviation, and the no-information allocation inherits this asymmetry. With three treatments, for example, the latter assigns 37 percent of the traffic to the control and 21 percent to each treatment.

In a stratified design, the population divides into segments or markets with known shares s_x , and the experimenter chooses the traffic of each segment and its treatment-control split. The loss

weights each segment by the square of its population share. Under the Neyman allocation, each treatment-by-segment cell receives traffic in proportion to $s_x \sigma_{dx}$, which favors the noisier segments. The no-information allocation instead samples each segment in proportion to its population share and splits it evenly between treatment and control.

In both designs the loss is a sum of variances divided by allocation shares, a confidence interval for each variance gives a rectangle, and the CMR allocation minimizes worst-case regret over it by solving a small convex program. The certificate and the guarantee of Theorem 1(i) carry over, and as the pilot grows the solution again moves from the no-information allocation toward the Neyman one.

	Two arms	Shared control, K treatments	S segments, shares s_x
Allocation	$\pi \in (0, 1)$	$(\pi_0, \dots, \pi_K), \sum_k \pi_k = 1$	$\pi_{dx}, \sum_{d,x} \pi_{dx} = 1$
Loss V	$\frac{\sigma_1^2}{\pi} + \frac{\sigma_0^2}{1-\pi}$	$\frac{K\sigma_0^2}{\pi_0} + \sum_{k=1}^K \frac{\sigma_k^2}{\pi_k}$	$\sum_{x=1}^S s_x^2 \left(\frac{\sigma_{1x}^2}{\pi_{1x}} + \frac{\sigma_{0x}^2}{\pi_{0x}} \right)$
Neyman	$\pi^* = \frac{\sigma_1}{\sigma_1 + \sigma_0}$	$\pi_0^* = \frac{\sqrt{K} \sigma_0}{\sqrt{K} \sigma_0 + \sum_j \sigma_j}, \pi_k^* = \frac{\sigma_k}{\sqrt{K} \sigma_0 + \sum_j \sigma_j}$	$\pi_{dx}^* = \frac{s_x \sigma_{dx}}{\sum_z s_z (\sigma_{1z} + \sigma_{0z})}$
No information	$\pi = 1/2$	$\pi_0 = \frac{1}{1 + \sqrt{K}}, \pi_k = \frac{1}{\sqrt{K}(1 + \sqrt{K})}$	$\pi_{dx} = s_x/2$

Table 1: The two-arm problem and its two platform extensions. In each column, the loss is proportional to the variance of the estimator, the Neyman allocation minimizes it when the variances are known, and the no-information allocation is what CMR returns when the confidence set is the whole variance space.

4 Simulations calibrated to two platform experiments

I compare the no-information allocation, trimmed FNA and CMR in two simulations calibrated to real platform experiments. The first has a shared control and the second is stratified. In each, I simulate binary outcomes at rates calibrated to the data, draw a pilot of between 32 and 4,000 observations, and let each rule set the main-wave allocation from that pilot. CMR uses the exact finite-sample intervals with $\alpha = 0.05$, the no-information allocation is the one in Table 1, and trimmed FNA keeps every arm’s share away from zero. The untrimmed rule assigns some arm zero traffic in 77 percent of Upworthy pilots of size 100 and in 99 percent of Criteo pilots, which makes its mean loss infinite at every pilot size. Performance is measured as $100 r/V^*$, the percent of additional main-wave traffic a rule needs in order to match the precision of the Neyman allocation.

The first experiment comes from Upworthy, where readers were randomly shown one of four images next to the same headline and the outcome is whether they clicked it [Matias et al., 2021]. Click rates differ little across the four images, which puts the no-information allocation within 1.3 percent of the Neyman allocation and leaves little to gain from adapting. The second is the Criteo uplift benchmark [Diemert et al., 2018], in which 13.98 million users were randomized to eligibility for an advertising campaign and the outcome is a visit to the advertiser’s site. I split users into four segments by the quartiles of one of their recorded characteristics. Visit rates range from 1.2 to 11.3 percent across the segments. The no-information allocation therefore needs 15.1 percent more traffic to match the Neyman allocation, which leaves much to gain from adapting to the pilot.

Figure 2 shows how performance varies with the pilot size for each rule. In Upworthy, CMR pays a small price for adapting, at most 2.2 percentage points above the no-information allocation over the pilot sizes reported, and by $M = 4,000$ it is back below it. Trimmed FNA does far worse, needing 183 percent more traffic than the Neyman allocation at $M = 100$ and still 57 percent more at $M = 500$. In Criteo, a pilot of 500 users lets CMR bring the excess traffic down from 15.1 to 5.1 percent, two thirds of what adapting can gain and roughly nine percent fewer main-wave users for

the same precision. With 4,000 pilot users the excess falls to 1.2 percent, 92 percent of the gain. Trimmed FNA still needs 73 percent more traffic at $M = 500$. Put simply, when there is little to find CMR costs almost nothing, and when there is much to find it finds most of it.

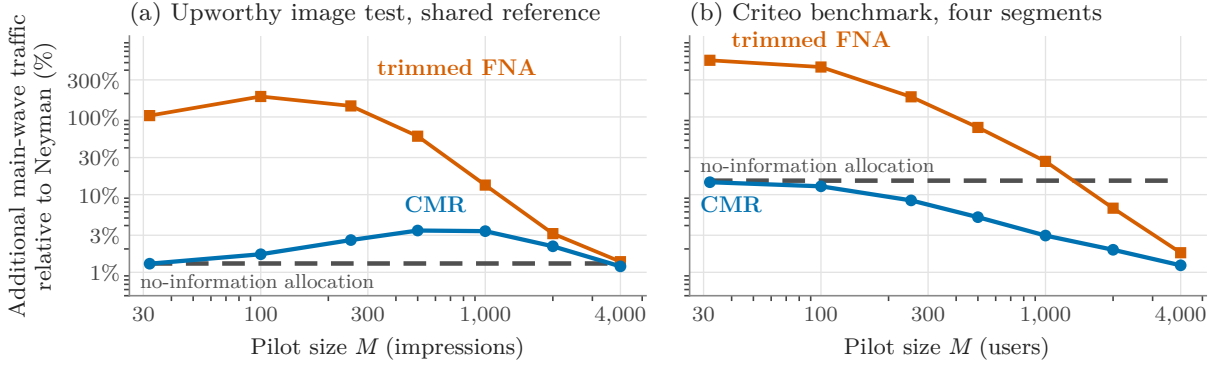


Figure 2: Efficiency loss by pilot size, i.e., the percent of additional main-wave traffic a rule needs to match the precision of the Neyman allocation. Both axes are in logs, and each point is the mean loss over 2,000 pilot draws. Pilot observations are excluded from the comparison. The no-information allocation is the shared-control allocation of Section 3 in panel (a) and representative sampling with even splits in panel (b). Trimmed FNA is the plug-in allocation with minimum shares of 5 percent per arm in (a) and a tenth of the representative share per cell in (b).

5 When to pilot, and how large

If a platform runs a ramp-up in any case, using it to set the split costs nothing extra, since its observations are collected anyway. A pilot run only to choose the split is different, because its observations are not part of the main wave. Suppose a total budget of n observations, of which M go to a balanced pilot. Moving from balance to the Neyman allocation saves at most a fraction $(\sigma_1 - \sigma_0)^2 / [2(\sigma_1^2 + \sigma_0^2)]$ of the sample, while the pilot uses up a fraction M/n . A pilot can therefore pay for itself only if

$$\frac{M}{n} \leq \frac{(\sigma_1 - \sigma_0)^2}{2(\sigma_1^2 + \sigma_0^2)}.$$

This condition is necessary rather than sufficient, since it assumes that the pilot learns the variances exactly. When it holds, the pilot has to be large enough to learn the variances yet small enough to leave traffic for the main wave. I show that a balanced pilot of roughly $n^{2/3}$ observations followed by CMR attains a worst-case regret within a constant factor of the smallest possible over all balanced pilot sizes and all assignment rules. For a budget of 1,000, that means a pilot of about 100 observations.

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